

Dartagnan Birnie

603-833-1781 · dartbirnie@gmail.com · dartbirnie.dev · linkedin.com/in/dartagnan-birnie · github.com/GetBirned

Alton, NH · Available for onsite work in NH and MA · **U.S. Citizen** · **Eligible for a U.S. Government security clearance**

SUMMARY

Software engineer with a B.S. in Computer Science and production experience across the full defect lifecycle — reproducing, root-causing, and verifying fixes in C#.NET and Microsoft SQL Server systems deployed to large multi-user environments. Builds and validates integrations against documented third-party interface standards, and has developed sensor-data applications ingesting GPS, IMU, and LIDAR telemetry. Seeking software or integration & test engineering work on mission-critical systems.

TECHNICAL SKILLS

Languages: C#, Python, Java, C, SQL, JavaScript / TypeScript, HTML5 / CSS, PHP

Platforms & Data: .NET Framework, .NET 8, Microsoft SQL Server, SSRS, SSMS, IIS, REST APIs, Node.js, React.js, PostgreSQL, MongoDB

Engineering Practice: Defect lifecycle management (JIRA, Salesforce), issue reproduction and fix verification, root-cause analysis, requirements documentation (BRDs), Agile / Scrum, Git, system deployment, upgrade, and environment configuration

Systems & Sensors: GPS / GNSS, IMU, and LIDAR data ingest, AEMP2 (ISO 15143-3) telematics integration, Raspberry Pi, OpenStreetMap / GIS, Windows Task Scheduler, batch scripting

EXPERIENCE

Trimble Inc. — B2W Software

Portsmouth, NH

Software Implementation Consultant

May 2026 – Present

- Lead end-to-end deployments of enterprise construction-estimating software built on the Microsoft .NET Framework and SQL Server — requirements capture, database build-out, configuration, integration, and customer acceptance.
- Build and validate machine-telematics integrations against the AEMP2 (ISO 15143-3) interface standard, connecting John Deere, Komatsu, and Samsara equipment data feeds to customer systems through REST APIs, scheduled jobs, and batch automation.
- Analyze SQL Server databases to verify data integrity, diagnose production defects, and tune workflows across multi-site deployments.
- Serve as the primary technical escalation point for licensing, activation, and provisioning across the B2W product suite, authoring the internal documentation and workflow improvements that standardized the team's escalation process.

Technical Support Engineer

Sept. 2025 – May 2026

- Resolved 1,400+ production issues in six months spanning SQL Server, authentication, licensing, reporting services (SSRS), IIS, cloud-hosted infrastructure, and third-party software integrations.
- Owned the defect lifecycle end to end: reproduced customer-reported failures in controlled test environments, isolated root cause from application logs and SQL Server data, documented and tracked defects in JIRA and Salesforce, and verified fixes with Development and QA prior to release.
- Maintained enterprise software environments through installations, version upgrades, IIS and SSRS configuration, database administration, cloud deployments, and application migrations.
- Promoted to Software Implementation Consultant within six months, recognizing technical performance and ownership of complex product workflows.

Account Tree

Dover, NH

Software Development & Marketing Intern

May 2024 – Aug. 2024

- Developed features and debugged PHP/JavaScript defects in a K-12 activity-fund platform, writing business requirements documents and working in an Agile cycle.

Designs By Dart LLC

Alton, NH

Founder / Web Developer

Jan. 2022 – Present

- Build and maintain sites for 20+ small-business clients; one of two web-development contractors approved by the NH SBDC and the State of New Hampshire.

PROJECTS

PiRail — Railroad Track Inspection Platform (UNH Senior Capstone)

Aug. 2024 – May 2025

Software Engineer · React, JavaScript, Python, Raspberry Pi, GPS / IMU / LIDAR

- Delivered the operator interface for a low-cost track-inspection system running on a Raspberry Pi with GPS, IMU, and LIDAR sensors — replacing a legacy jQuery interface with a mobile-first React application that plots live position and lets crews report track points of interest.
- Built a playback simulator that replays recorded sensor runs, letting the team test and validate the interface against real captured telemetry without hardware time on the rail.
- Coordinated a five-person team across two semesters using Git and Agile practices, contributing to an active open-source repository (cpn18/track-chart).

DartERP — Manufacturing ERP

Personal Project

C#, .NET 8, WinForms, SQL Server

- Built a desktop manufacturing ERP for a firearms manufacturer — inventory, purchase orders, work orders, and serialized finished-goods tracking with full audit trails, modeled on regulated-manufacturing traceability requirements.
- Designed the SQL Server schema and data-access layer, structuring the app in a layered .NET architecture.

EDUCATION

University of New Hampshire — B.S. in Computer Science · Dean's List: Fall 2024, Spring 2025

Durham, NH · May 2025